

# Sanando Banerjee

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## PROFESSIONAL SUMMARY

Software Engineering undergraduate specializing in backend and full-stack development, with hands-on experience building scalable APIs, real-time applications, AI-powered systems, and data-driven platforms. Proficient in Python, TypeScript, databases, cloud technologies, and modern development tools, with a strong foundation in data structures, distributed systems, and software engineering principles.

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## EDUCATION

Meghnad Saha Institute of Technology

Bachelor of Technology (B. Tech) in Information Technology

Kolkata, India

Aug 2023 - Jul 2027

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## TECHNICAL SKILLS

**Programming:** Python, C++, TypeScript, JavaScript

**Frontend:** Next.js, React

**Backend & APIs:** FastAPI, REST APIs, Server-Sent Events

**AI/ML:** LangChain, LangGraph, RAG, Groq, MCP

**Databases:** MongoDB, ChromaDB, SQL Server

**Testing:** Pytest, Playwright

**DevOps & Cloud:** Git, Docker, CI/CD, Linux, AWS, Azure

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## RELEVANT EXPERIENCE

**bITsera – Tech Club of IT, MSIT | Web Development Lead**

Led and organized React workshops for 60+ students, covering state management, API integration and React hooks while emphasizing the importance of modular architecture and reusable component design.

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## PROJECT EXPERIENCE

### Saturn-AI – Full-Stack AI Trading Intelligence Platform

- Engineered a full-stack Bitcoin intelligence platform that ingests live cryptocurrency news from **5 RSS feeds** every **10 minutes**, leveraging **Groq**, **FastAPI**, and **MongoDB Atlas** to generate AI-powered **BUY/SELL/HOLD** trading signals with confidence scoring.
- Designed a modular backend following **SOLID principles**, implementing **9+ REST APIs**, **APScheduler**-based recurring pipelines, a streaming AI explanation service, and an **MCP server** exposing trading tools to Claude Desktop and other MCP-compatible AI agents.
- Developed a **React** dashboard delivering real-time sentiment feeds, Bitcoin price tracking via **CoinGecko**, streaming AI market analysis, and interactive visualizations through low-latency backend services.

### **DDoS Overwatch – Real-Time Cyberattack Visualization Platform**

- Built a real-time DDoS attack visualization dashboard using **Next.js**, **React**, **Zustand** and **Server-Sent events** to stream simulated attack telemetry across 13 geographic regions with **sub-second UI update latency**.
  - Designed a **modular event pipeline** – SSE producer, buffered in-memory store, geospatial React visualisation – modelling randomized attack type, bandwidth (1-100 Mbps) and duration data across a 1,000-event rolling buffer.
  - Implemented centralized **Zustand state management** to keep client-side visualization synchronized with a continuous simulated event stream.
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